

A COUNTEREXAMPLE TO THE LAU–SHIU–NG CONJECTURE ON MÖBIUS LADDERS

ABSTRACT. We give a local antimagic labeling of the eight-vertex Möbius ladder M_8 with three induced colors and prove that $\chi_{\text{la}}(M_8) = 3$. This refutes Conjecture 3.14 of Lau, Shiu, and Ng at its first admissible parameter. The same labeling also gives a counterexample to Theorem 2.4 of arXiv:2203.06594v1, published as Acta Math. Hungar. **169** (2023), 108–133.

1. THE COUNTEREXAMPLE

Let G be a graph with q edges. A *local antimagic labeling* is a bijection

$$f: E(G) \longrightarrow \{1, \dots, q\}$$

such that the induced vertex sums

$$f^+(x) := \sum_{e \ni x} f(e)$$

satisfy $f^+(x) \neq f^+(y)$ whenever $xy \in E(G)$. The local antimagic chromatic number $\chi_{\text{la}}(G)$ is the minimum number of distinct induced vertex sums over all local antimagic labelings of G .

For $n \geq 2$, let M_{2n} be obtained from the cycle

$$u_1 u_2 \cdots u_n v_1 v_2 \cdots v_n u_1$$

by adding the edges $u_i v_i$ for $1 \leq i \leq n$. Lau, Shiu, and Ng conjectured that $\chi_{\text{la}}(M_{2n}) = 4$ for every even $n \geq 4$ [1, Conjecture 3.14]. The first case covered by the conjecture is M_8 .

Theorem 1. *The Möbius ladder M_8 satisfies*

$$\chi_{\text{la}}(M_8) = 3.$$

Consequently, Conjecture 3.14 of Lau, Shiu, and Ng is false.

Proof. Define $f: E(M_8) \rightarrow \{1, \dots, 12\}$ by

$u_1 u_2 \mapsto 1$	$u_2 u_3 \mapsto 8$	$u_3 u_4 \mapsto 3$
$u_4 v_1 \mapsto 7$	$v_1 v_2 \mapsto 5$	$v_2 v_3 \mapsto 2$
$v_3 v_4 \mapsto 11$	$v_4 u_1 \mapsto 4$	$u_1 v_1 \mapsto 12$
$u_2 v_2 \mapsto 10$	$u_3 v_3 \mapsto 6$	$u_4 v_4 \mapsto 9$

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The labels are exactly $1, \dots, 12$. Direct calculation gives

x	u_1	u_2	u_3	u_4	v_1	v_2	v_3	v_4
$f^+(x)$	17	19	17	19	24	17	19	24.

Thus the fibers of f^+ are

$$\{u_1, u_3, v_2\}, \quad \{u_2, u_4, v_3\}, \quad \{v_1, v_4\},$$

with induced colors 17, 19, and 24, respectively. Each fiber is an independent set in M_8 , so f is a local antimagic labeling and $\chi_{\text{la}}(M_8) \leq 3$.

The vertices u_1, u_2, u_3, u_4, v_1 form a 5-cycle. Hence $\chi(M_8) \geq 3$. Since every local antimagic labeling induces a proper vertex coloring,

$$\chi_{\text{la}}(M_8) \geq \chi(M_8) \geq 3.$$

Therefore $\chi_{\text{la}}(M_8) = 3$. Since $M_8 = M_{2n}$ for the admissible parameter $n = 4$, the conjecture is false. $\square$

Remark. Theorem 2.4 of arXiv:2203.06594v1 reads, in the notation used here, “Suppose G is a cubic graph of order 8, then $\chi_{\text{la}}(G) = 4$ ” [2]. It stands in a subsection titled “Connected Cubic Graphs of Order at Most 8”, and its proof reduces the claim to the four graphs M_8, H_1, H_2, H_3 enumerated there, so the intended scope is the connected cubic graphs of order 8. Since M_8 is a connected cubic graph of order 8 (indeed the first of those four graphs), the theorem above also disproves that statement. That proof uses

$$2x + 3y + 3z = 132$$

for a hypothetical three-coloring with color-class sizes 2, 3, 3. For any labeling of a 12-edge graph by $1, \dots, 12$, however,

$$\sum_{v \in V(G)} f^+(v) = 2 \sum_{e \in E(G)} f(e) = 2 \sum_{i=1}^{12} i = 156.$$

The value 132 equals $2 \sum_{i=1}^{11} i$, so the labels were summed as if there were eleven of them rather than twelve. With the correct total, the last of the five pairs $(x, \{y, z\})$ that the source derives from its identity, namely $(12, \{17, 19\})$, becomes $(24, \{17, 19\})$, and this is the pair realized by the labeling above:

$$2 \cdot 24 + 3 \cdot 17 + 3 \cdot 19 = 156.$$

The quotation and the identity above are taken from the v1 e-print; we have not compared them with the corresponding text of the published version. The forward citations of [1] and of arXiv:2203.06594 were surveyed; the value $\chi_{\text{la}}(M_8) = 3$ does not appear among them, and no erratum to either source is known to us. No claim is made here about $\chi_{\text{la}}(M_{2n})$ for even $n \geq 6$.

REFERENCES

- [1] G.-C. Lau, W.-C. Shiu, and H.-K. Ng, *On local antimagic chromatic number of cycle-related join graphs*, Discuss. Math. Graph Theory **41** (2021), 133–152. doi:10.7151/dmgt.2177.
- [2] G.-C. Lau and W.-C. Shiu, *On join product and local antimagic chromatic number of regular graphs*, arXiv:2203.06594v1, 2022; published as Acta Math. Hungar. **169** (2023), 108–133, doi:10.1007/s10474-023-01298-7.